

Salam Afouni

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Education

Columbia University

New York, USA

PhD in Operations Research

Sept. 2023 - Present

- **Advisor:** Prof. Christian Kroer
- **Relevant Courses:** Algorithmic Game Theory, Optimization, Market Design, Network Games, Algorithm Analysis, Stochastic Modeling

American University of Beirut

Beirut, Lebanon

BE in Industrial Engineering, OR track

Aug. 2019 - May 2023

- Minor in Applied Mathematics
- Graduated with High Distinction. Cumulative GPA **4/4**
- **Relevant Courses:** Operations Research, Advanced Optimization Techniques, Numerical Computing, Numerical Linear Algebra, Discrete Events Simulation, Algorithmic Graph Theory, Data Analytics for Operations Research

Publications

Afouni, S., Černý, J., Ling, C. K., & Kroer, C. (2025). Commitment to Sparse Strategies in Two-Player Games. AAAI 25 📄 github.com/CoffeeAndConvexity/SparseEquilibria

Afouni, S., Černý, J., Ling, C. K., & Kroer, C. (2026). Colonel Blotto with Battlefield Games. AAAI26 📄 github.com/CoffeeAndConvexity/ColonelBlottoWithBattlefieldGames

Pacing Equilibrium for Abstract Mechanisms

Under Review

Smooth Bidding in First-Price Auctions with Predictions

Under Review

Experience

Research Assistant

New York, USA

Coffee and Convexity Lab - Columbia University

Sept. 2023 - Present

- **Research Interests:** Game Theory, Optimization, Mechanism Design, Economics and Computation
- **Specific:** Equilibrium Computation, Auctions Design, Optimization over Networks, Online Learning, Market Equilibrium

Visiting Researcher

Paris, France

FairPlay Team - CREST, ENSAE

Feb. 2026 - June 2026

- Collaborated with Prof. Vianney Perchet and Dr. Simon Mauras on projects in mechanism design and auction theory
- **Academic areas:** Mechanism Design with Predictions, Market Design, Auction Theory

Undergraduate Research Assistant

Beirut, Lebanon

Control and Optimization Laboratory - American University Beirut

Oct. 2022 - Sept 2023

- Worked with Prof. Dany Abou Jaoude on the problem of distributed resource allocation over graphs with uncertainty
- Developed a distributed algorithm for fair resource allocation with improved performance under constant step-size
- **Academic areas:** Distributed Optimization, Resource Allocation

Research Intern - EPFL Excellence in Engineering Fellowship Program

Lausanne, Switzerland

Adaptive Systems Laboratory - EPFL

July. 2022 - Sept. 2022

- Worked under the supervision of Prof. Ali H. Sayed and Dr. Flavio Pavan on the problem of Optimization Over Competing Networks
- Formulated and analyzed the local and global distributed optimization problems in the LMS setting
- Studied the relationship between Nash equilibrium and Pareto optimal solutions for the multi-objective optimization problem
- **Academic areas:** Adaptive and Distributed Optimization, Game Theory

Talks and Presentations

2026 **Workshop Talk**, GAIMSS'26

Bonn, Germany

2026 **Workshop Talk**, MIW'26, Pacing Equilibrium for Abstract Mechanisms

Online

2024 **Tutorial Presentation**, IEOR PhD Tutorial, Intro to Auctions and Pacing Equilibria in Auction Markets

IEOR dept.

Teaching

IEOR 4004 Optimization Models and Methods
CSOR 4231 Analysis of Algorithms

Fall 2025, Spring 2026
Spring 2025

Projects

Columbia University

IEOR Department

- Coalition Formation in Auctions via Social Learning
- The Connection between Stable Matching and Optimal Transport

American University of Beirut

Industrial Engineering Department

- Machine Learning for Crowdfunding Success
- Queuing System Analysis for Optimal Server Allocation at Checkouts in Spinneys Supermarket, Lebanon
- Gradient Descent Convergence Guarantee by the Stable Manifold Theorem
- Movie Recommendation System
- Cost Optimization: A Poisson Process and A Markov Chain Approach

Honors and Awards

2023	LebNet EiC Scholarship , LebNet Program	USA
2019-2023	Dean's Honor List , American University of Beirut	Lebanon
2020-2023	Francois Hamze Scholarship , American University of Beirut	Lebanon
2022	EPFL Excellence in Engineering Fellowship , EPFL	Switzerland

Service and Outreach

2026 **Program Committee member**, AAAI'27
2025 **Reviewer**, AAAI'26
Present **Member of the IEOR PhD Council**, IEOR Department

Skills

Programming Languages	Python, MATLAB, R
Numerical Libraries	Gurobi, CVXPY, Matplotlib, NumPy, SciPy, pandas
Soft Skills	Academic writing, Presentation skills, Leadership skills, Hands-on learning

Languages

English	Professional proficiency	IELTS 8
French	Bilingual Proficiency	
Arabic	Native Proficiency	